CDI COLLEGE • DATA SCIENCE TOOLS AND TECHNIQUES

Technical Analysis Using R

ASSIGNMENT 02 / PRELIMINARY STAGE

Stock portfolio ingestion, descriptive analysis, 20-day moving averages, and reproducible outputs.

PROJECT DETAIL	CONFIRMED VALUE
Student	Mohammad Aljarad
Course	CA-BDA400 / Data Science Tools and Techniques
Assessment	Assignment 2 / 5%
Date of execution	October 31, 2026
Portfolio	AAPL • MSFT • GOOGL
Source	RStudio 4.6.2 / quantmod, TTR and zoo

Executive summary

The preliminary stage was executed locally using RStudio and source scripts saved under the Assignment2 directory. All three stock symbols were processed successfully. Each symbol produced a daily data export and a chart comparing closing prices with a 20-day simple moving average. A consolidated CSV contains the calculated descriptive statistics.

The student created and updated a public GitHub repository, TechnicalAnalysis. Screenshots taken during the actual workflow and the original generated charts are reproduced in this report. Values in the results tables are taken from the submitted statistics_summary.csv, not estimated from the charts.

Project repository: github.com/maljarad30-coder/TechnicalAnalysis

Evidence note: The RStudio installation itself was performed before these screenshots were captured. The available image confirms an operational RStudio / R environment, but not every original installer screen.

01 / OBJECTIVE AND REPOSITORY SETUP

Purpose and assessed deliverables

The assignment requires a GitHub repository, a text portfolio of ticker symbols, R scripts that import and summarize stock data, visual outputs, and a step-by-step report with screenshot evidence.

DELIVERABLE	STATUS / EVIDENCE
TechnicalAnalysis repository	Created publicly; commits captured in GitHub screenshots.
Assignment2 subdirectory	Contains scripts, portfolio, report and output folders.
portfolio.txt	Three ticker symbols, one per line: AAPL, MSFT, GOOGL.
R source code	Separate package, import/statistics and display/plot scripts.
Generated results	3 data CSVs, 1 summary CSV, and 3 PNG charts confirmed locally.
Report and screenshots	English report with authentic user-provided evidence.

GitHub repository creation

The repository was created under the student-owned GitHub account and made public, with README.md initialized. The next step was to upload an Assignment2 subdirectory, preserving a distinct folder for this course assignment.

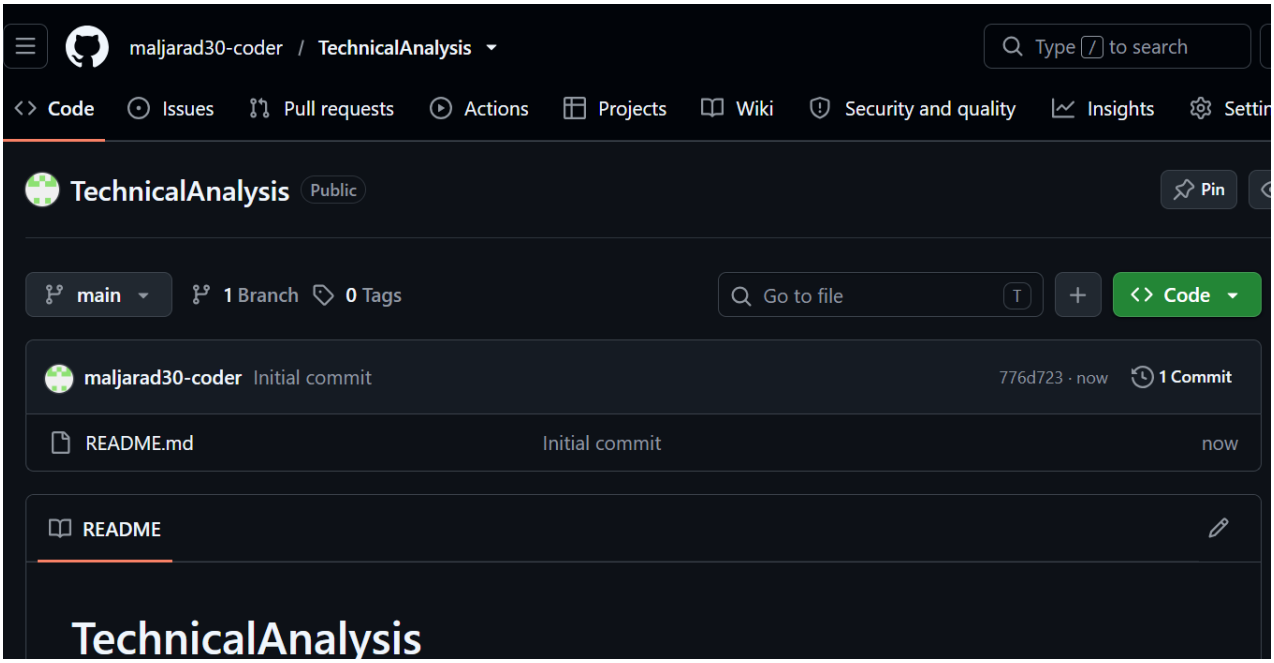


Figure 1. GitHub repository created with the required TechnicalAnalysis name.

02 / UPLOADED CODE AND FOLDER LAYOUT

Repository contents and commits

The uploaded folder preserves the R code, reporting documents, portfolio list, and designated outputs/screenshots directories. The GitHub upload view below records the required source files under the Assignment2 prefix.

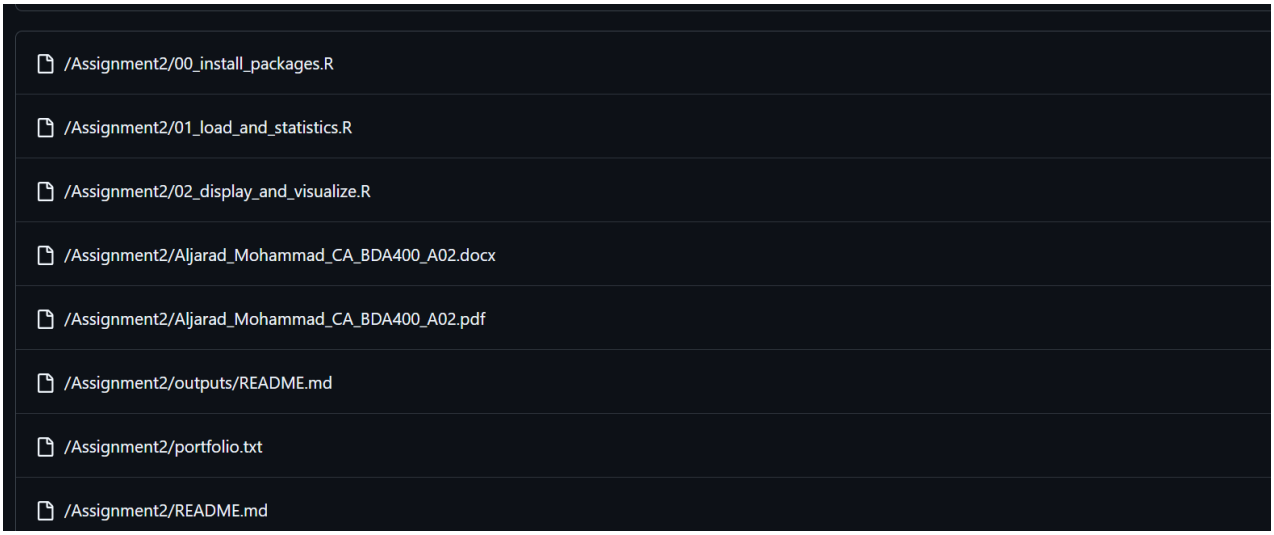


Figure 2. Actual GitHub upload list showing the Assignment2 project files.

After uploading the source materials, a further GitHub commit was made to add generated results. The final repository screenshot records three commits; it establishes that an upload succeeded, but should not substitute for checking individual result files in the repository.

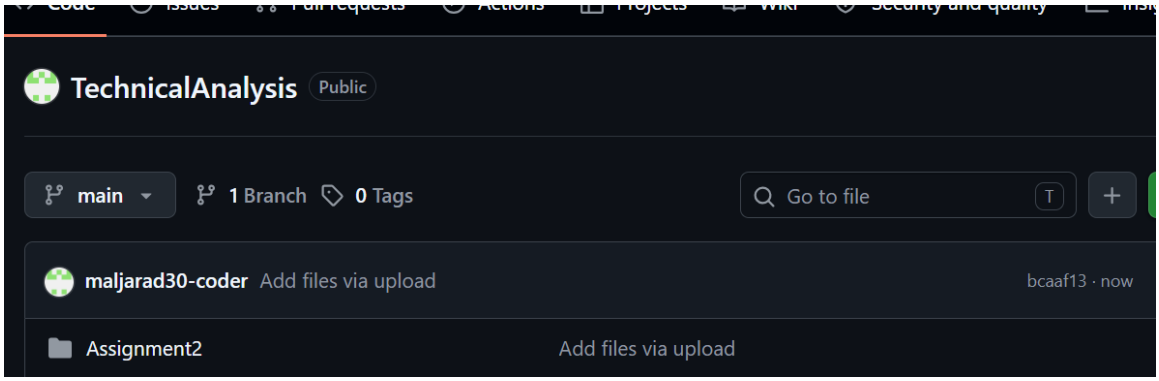


Figure 3. GitHub repository after the third commit; Assignment2 is present.

03 / ENVIRONMENT AND PACKAGE VERIFICATION

R / RStudio installation state

RStudio was launched successfully on a Windows computer. Its Console shows R version 4.6.2, and the project was run from the Assignment2 working directory. This is verification of a working installation, not a reconstructed screenshot of the download or installation wizard.

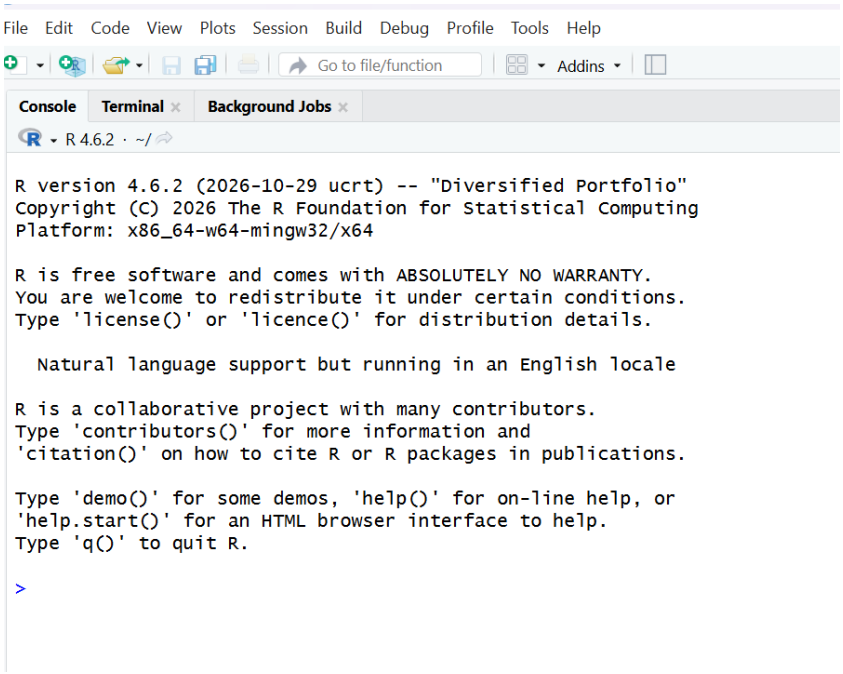


Figure 4. RStudio launched with an active R 4.6.2 Console.

Required packages installed

The source file 00_install_packages.R checks whether quantmod, TTR and zoo are present. The student then verified their namespaces using requireNamespace(..., quietly = TRUE). All three returned TRUE in the actual Console.

```
All required packages are installed: quantmod, TTR, zoo
R version: R version 4.6.2 (2026-10-29 ucrt)
> sapply(c("quantmod", "TTR", "zoo"), requireNamespace, quietly = TRUE)
quantmod      TTR      zoo
    TRUE     TRUE     TRUE
> sapply(c("quantmod", "TTR", "zoo"), requireNamespace, quietly = TRUE)
quantmod      TTR      zoo
    TRUE     TRUE     TRUE
>
```

Figure 5. Console evidence that quantmod, TTR and zoo each returned TRUE.

04 / PORTFOLIO AND ANALYTICAL FUNCTIONS

Input file and implementation

The project uses portfolio.txt as a plain-text ticker list, with one stock symbol per line:

AAPL
MSFT
GOOGL

The script 01_load_and_statistics.R defines reusable functions. It validates symbols, retrieves daily OHLCV stock prices through quantmod, constructs one data frame per symbol, calculates statistics, and returns a consolidated structured summary.

FUNCTION	ROLE
read_portfolio()	Reads, trims, uppercases and validates ticker symbols.
load_stock_data()	Downloads price data and stores a separate data frame for each stock.
rounded_price_mode()	Returns all tied modal closing prices after rounding to cents.
calculate_statistics()	Calculates latest close, mean, median, mode, sample SD and 20-day SMA.
run_analysis()	Runs the complete three-stock workflow and creates a combined summary.

Statistical calculation definitions

- Mean and median: central tendency of the daily closing-price observations.
- Mode: price values rounded to two decimals; tied values are displayed together when no single mode dominates.
- Standard deviation: sample standard deviation of the valid daily closing-price observations.
- SMA20: trailing arithmetic average of closing prices over the preceding 20 sessions; the first 19 values have no complete window.

The source file was loaded into RStudio successfully, and the defined functions appeared in the Environment pane.

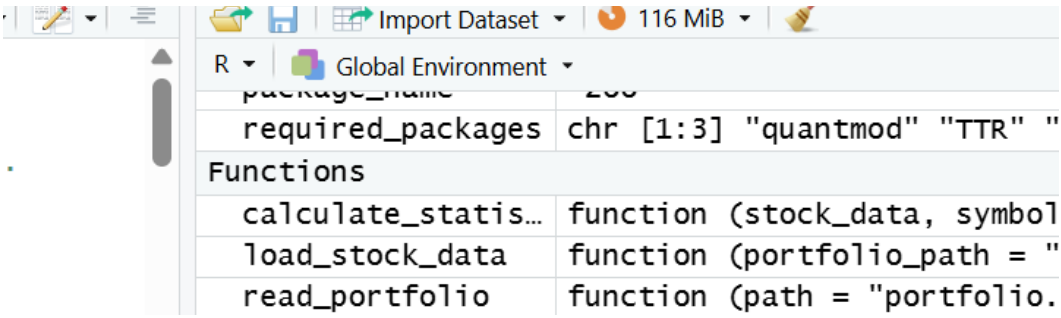


Figure 6. The defined stock-analysis functions in RStudio Environment.

05 / OBSERVED STATISTICAL RESULTS

Descriptive statistics from the executed R script

The uploaded statistics_summary.csv reports 251 observations for each ticker, spanning 2025-10-31 through 2026-10-30. All prices below are in USD, rounded to two decimal places.

Symbol	Sessions	Latest	Mean	Median	Std dev	SMA20
AAPL	251	344.63	291.56	283.10	29.62	342.08
MSFT	251	521.91	448.05	451.10	52.43	530.98
GOOGL	251	279.84	333.78	336.02	28.97	333.91

Modes of daily closing prices

AAPL: 272.95; 278.78; 302.25; 316.22; 333.02

MSFT: 390.34; 481.63; 483.47; 496.82

GOOGL: 318.58; 320.21; 337.12; 387.66

The ModeClose field contains multiple tied modes (rather than a single number), because the R function rounds prices to cents and returns every value tied at the maximum observed frequency. This is not the most frequent stock symbol or the statistical mode of raw market volume.

Latest close compared with SMA20

- AAPL: latest close $344.63 was above its 20-day SMA $342.08 by $2.55.
- MSFT: latest close $521.91 was below its 20-day SMA $530.98 by $9.07.
- GOOGL: latest close $279.84 was below its 20-day SMA $333.91 by $54.07.

Successful execution

The display script called run_analysis(), printed its summary, exported CSV files and PNG charts, and concluded with a success message in the actual RStudio Console.

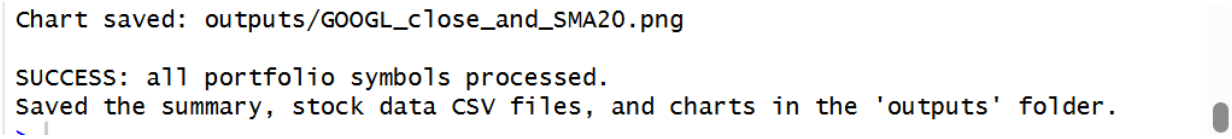


Figure 7. Actual RStudio Console showing successful processing and saved outputs.

06 / OUTPUT FILES AND REPRODUCIBILITY

Actual generated output files

The File Explorer capture below documents the seven outputs generated on October 31, 2026. All output files are organized under Assignment2/outputs.

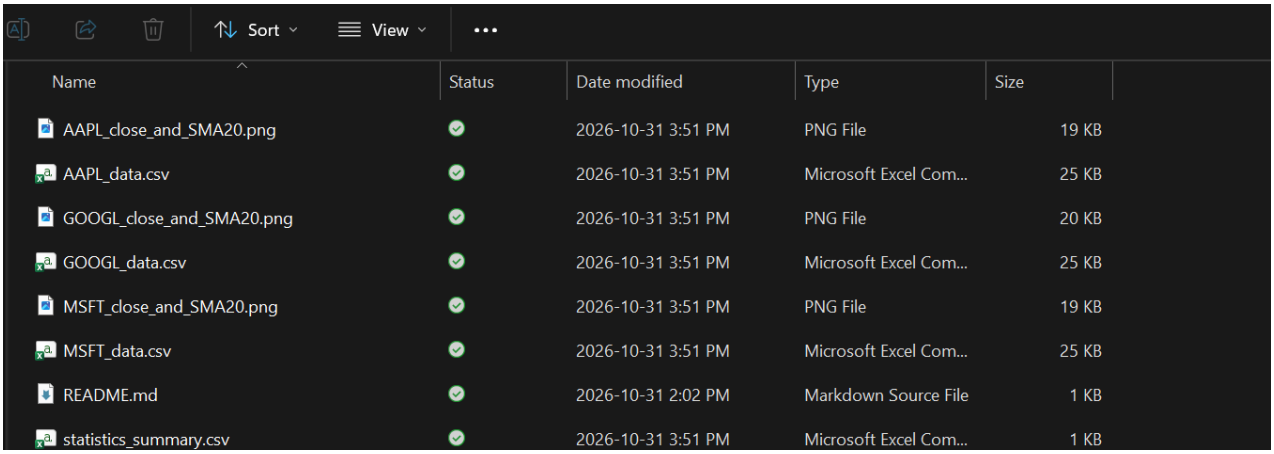


Figure 8. File Explorer listing the seven generated CSV/PNG analytical files.

OUTPUT TYPE	FILES
Data exports	AAPL_data.csv, MSFT_data.csv, GOOGL_data.csv
Statistics export	statistics_summary.csv
Charts	AAPL_close_and_SMA20.png; MSFT_close_and_SMA20.png; GOOGL_close_and_SMA20.png

Reproduce the results

1. Open RStudio and set its working directory to Assignment2. 2. Run 00_install_packages.R, then verify the three required packages. 3. Run 02_display_and_visualize.R, which sources 01_load_and_statistics.R. 4. Verify the final SUCCESS message and inspect the outputs subfolder.

Because the script requests approximately one year of observations relative to Sys.Date(), future runs will produce different sample windows and prices. The data also depend on Yahoo Finance connectivity and its historical records.

07 / STOCK VISUALIZATION: AAPL

AAPL | Daily close versus 20-day SMA

Time series for AAPL, covering October 31, 2025 to October 30, 2026. The blue line depicts daily closing prices and the orange line shows a trailing 20-session simple moving average.

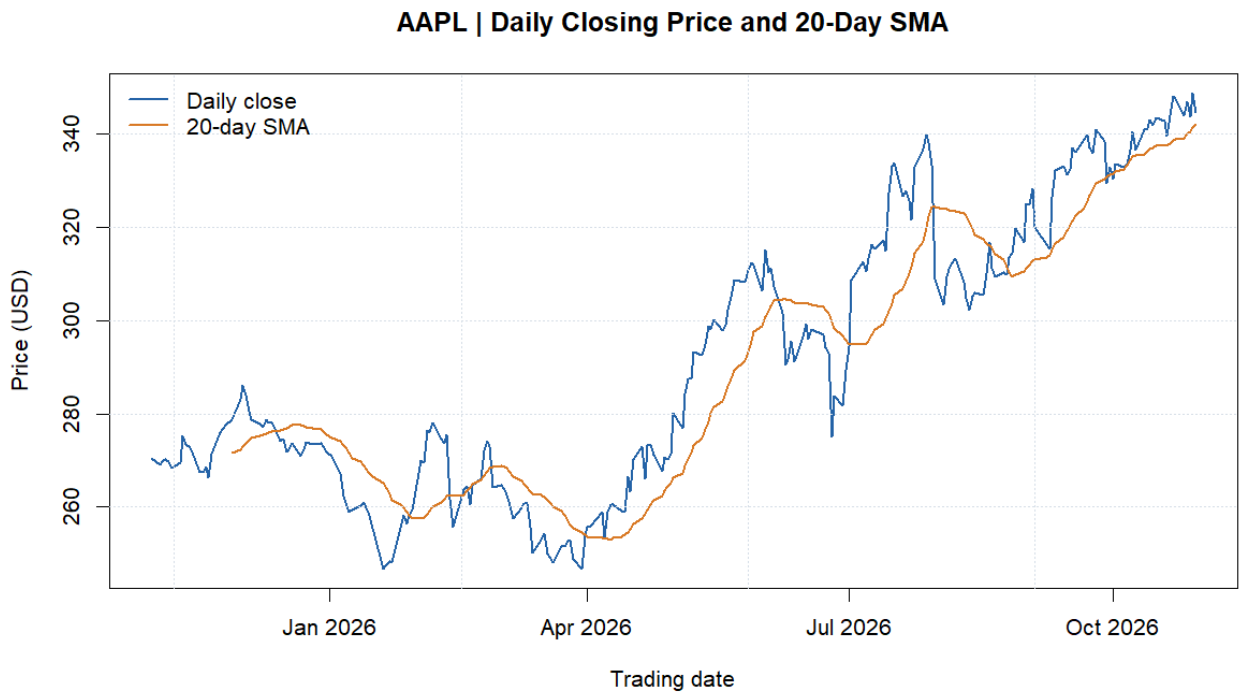


Figure 9. AAPL generated closing-price and 20-day SMA chart (original R output).

Interpretation of the observed chart

The final recorded close was $344.63; the corresponding 20-day moving average was $342.08. Therefore, the closing price ended above the smoothed average, with a difference of $2.55.

The series shows price fluctuations early in the sample and a stronger rise in the later months. The 20-day line is smoother and reacts to price changes after they occur.

This interpretation is descriptive only and is not an investment forecast or trading recommendation.

08 / STOCK VISUALIZATION: MSFT

MSFT | Daily close versus 20-day SMA

Time series for MSFT, covering October 31, 2025 to October 30, 2026. The blue line depicts daily closing prices and the orange line shows a trailing 20-session simple moving average.

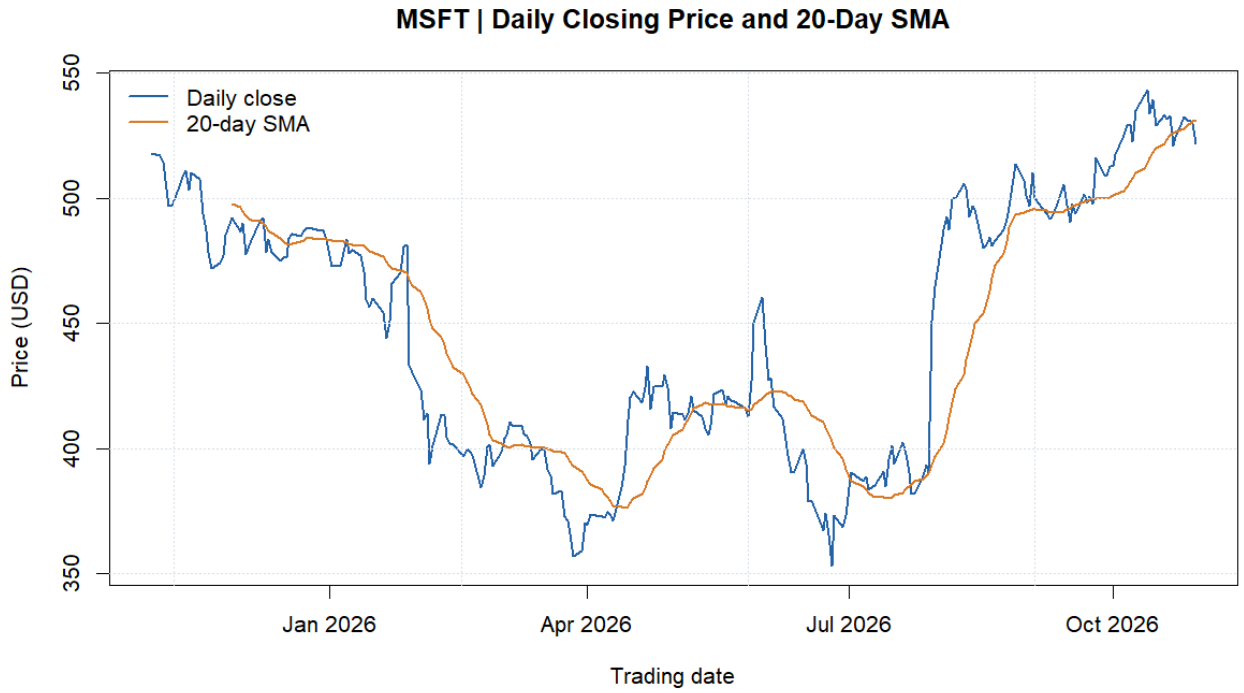


Figure 10. MSFT generated closing-price and 20-day SMA chart (original R output).

Interpretation of the observed chart

The final recorded close was $521.91; the corresponding 20-day moving average was $530.98. Therefore, the closing price ended below the smoothed average, with a difference of $9.07.

The series fell over the earlier part of the sample and later recovered. At the last observation, the close lies marginally beneath the moving-average line.

This interpretation is descriptive only and is not an investment forecast or trading recommendation.

09 / STOCK VISUALIZATION: GOOGL

GOOGL | Daily close versus 20-day SMA

Time series for GOOGL, covering October 31, 2025 to October 30, 2026. The blue line depicts daily closing prices and the orange line shows a trailing 20-session simple moving average.

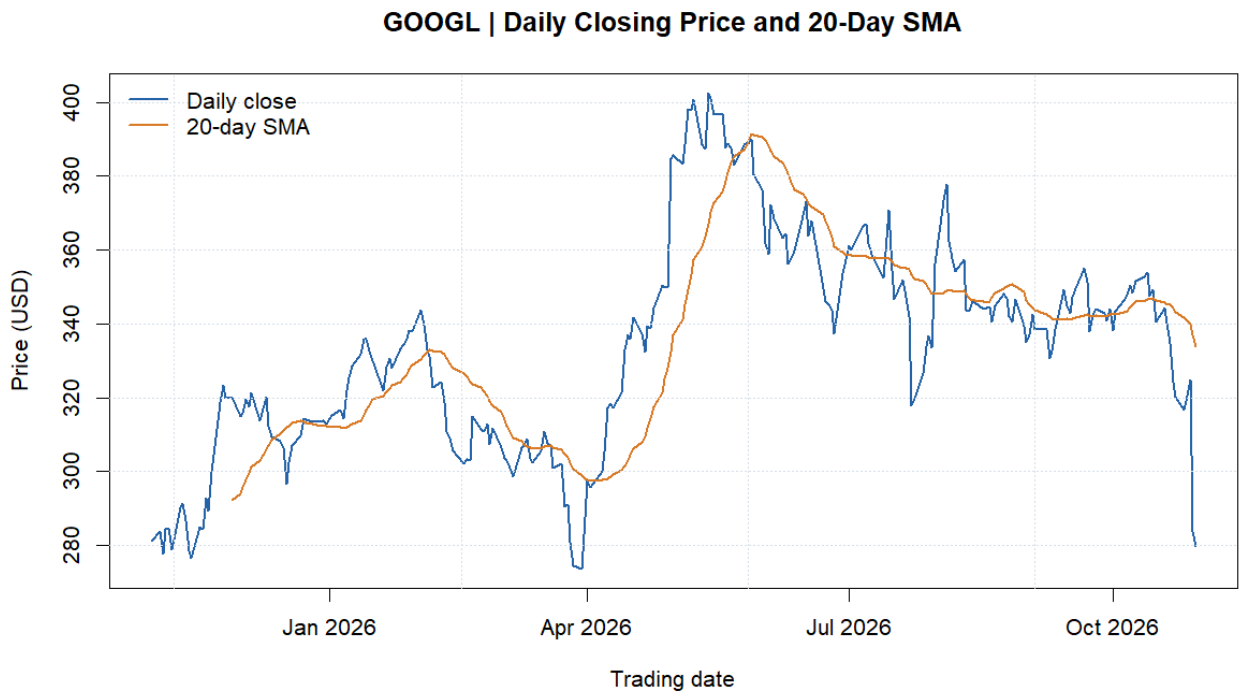


Figure 11. GOOGL generated closing-price and 20-day SMA chart (original R output).

Interpretation of the observed chart

The final recorded close was $279.84; the corresponding 20-day moving average was $333.91. Therefore, the closing price ended below the smoothed average, with a difference of $54.07.

The series shows a mid-sample peak followed by volatile and weaker closes late in the year. The final close is materially below the moving-average line.

This interpretation is descriptive only and is not an investment forecast or trading recommendation.

10 / SUBMISSION RECORD AND EVIDENCE INVENTORY

Completion and evidence record

ASSESSMENT ITEM	EVIDENCE PRESENT IN THIS REPORT
GitHub account and repository	Repository creation and final-commit screenshots.
R / RStudio setup	Working RStudio session and R version verification.
Package installation	Actual quantmod / TTR / zoo verification: TRUE / TRUE / TRUE.
Portfolio configuration	portfolio.txt content documented and file included in project.
Stock data functions	Function names confirmed in RStudio Environment.
Descriptive statistics	Exact statistics_summary.csv values and method definitions.
Data display and plots	Three original charts and File Explorer output inventory.
Documentation	This step-by-step English report, real workflow screenshots and repository reference.

Scope of available screenshot evidence

The screenshots support a successful operating environment and completed analytical run. The original installation wizard screens and an open portfolio.txt screenshot were not provided; this report does not claim they were captured. If requested by the instructor, these can be supplemented with new setup or file-opening evidence, clearly dated as such.

Repository and submission

Student GitHub link: https://github.com/maljarad30-coder/TechnicalAnalysis

Repository folder: TechnicalAnalysis/Assignment2. Upload this revised Word document, the revised PDF, and the evidence screenshot files into the existing Assignment2 folder. If LMS accepts a PDF upload, submit the PDF and provide the repository link on its cover page or in the submission comments. The instructor PDF separately specifies Word in GitHub.

References

- CDI College. (2026). BDA400 Assignment 2: Technical Analysis Using R, Preliminary Stage (instructor handout).
- R Foundation for Statistical Computing. R 4.6.2 (installed and observed in RStudio).
- quantmod, TTR and zoo packages: CRAN package documentation.
- Source of observed results: student-generated statistics_summary.csv and three generated PNG charts, October 31, 2026.